

Hi Translate: Breaking down language barriers with real-time translation tools

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ABOUT HI TRANSLATE



With the artificial intelligence and machine learning technology on Google Cloud, Hi Translate enhances the performance of its translation applications to support instant translation of text, voice, and image across more than 100 languages.

About Hi Translate

Hi Translate is a startup focusing on translation and language education applications. Aiming to bring the best possible translation tools to connect people around the world, Hi Translate has released two products offering cross-application real-time translation of text, voice, and image. It now has more than three million daily active users across 210 multilingual countries.

Industries: Technology

Google Cloud results

- Increases user retention by 20% with high-quality translations in 108 languages
- Enables identification of multiple languages in one image using Cloud Vision API to simplify the translation process
- Improves connection stability with a highly connected global network, ensuring zero downtime in more than six months

Location: Hong Kong

Shorter
translation
process
by 40%

Translation API

Google Cloud (<https://cloud.google.com/>)

Cloud SDK (<https://cloud.google.com/sdk/>)

Cloud Vision API (<https://cloud.google.com/vision/>)

Cloud Speech to Text
(<https://cloud.google.com/speech/>)

Cloud Translation API
(<https://cloud.google.com/translate/>)

- of
deployment
- Frees up
DevOps
workforce to
focus on
product
development
by reducing
maintenance
team from
three to one

Imagine living in a world without language barriers. People text each other in different languages without worrying about being misunderstood. Taxi drivers instantly know where foreign travelers want to go. And strangers can have conversations despite speaking different languages.

That's the vision of Li Yang, CEO at [Hi Translate](https://play.google.com/store/apps/details?id=com.zaz.translate)
(<https://play.google.com/store/apps/details?id=com.zaz.translate>)

, since the day he co-launched the first version of the company's translation application Hi Translate in early 2018. Since then, he and his partner have been striving to enhance user experience by providing more accurate translations conveniently. Hi Translate is one of the first translation apps worldwide supporting cross-application translation, which enables users to acquire instant translations directly in chat apps or on web pages.

"For a lean team of 10, it would have been too time consuming and resource intensive to improve our services solely on our own. Google Cloud helped us overcome these roadblocks with its advanced artificial intelligence (AI) and machine learning (ML) technologies."

—Li Yang, CEO, Hi Translate

Besides translation services, Hi Translate as a company also hopes to help its users build up their language skills. In mid-2018, it released [Hi](#)

Dictionary

(<https://play.google.com/store/apps/details?id=com.talpa.translate>)

, an education application that combines dictionary, translation, and proofreading features. The company has grown in popularity in a number of multilingual countries, notably India, Bangladesh, and Nigeria due to the high usability of its product. As of December 2020, the company's two applications have more than 30 million downloads, with over three million daily active users relying on its services for everyday communication.

As a part of its expansion strategy, Hi Translate plans to increase the number of the languages supported by its apps. As most of Hi Translate's users live in regions with frequently interrupted internet connection, one of its priorities is to ensure the stability of its services no matter where the users are. To achieve these two goals, the company conducted a series of A/B tests, wherein two versions of its app are pitted against each other on different cloud providers, and eventually decided to migrate its translation business to Google Cloud (<https://cloud.google.com/>) in May 2020.

“Among all the
products we
evaluated, Cloud
Translation API

provides the
best translation
quality and
supports the
most languages
across our
target markets.
We are therefore
able to greatly
optimize our
translation
applications and
reach wider
users.”

—Li Yang, CEO, Hi
Translate

Supporting more than 100 languages with high-quality translation

As with any translation software, the quality of translation is the primary factor that affects user satisfaction. According to Hi Translate’s A/B test result, the user retention rate is 20% higher on the translation app powered by Google Cloud compared to other alternatives, which made the team confident of the move.

Previously, Hi Translate relied on another cloud provider, third-party APIs, and its own technologies to support its translation services. But there was still room for improvement in terms of the translation quality and the variety of languages. “For a lean team of 10, it would have been too time consuming and resource intensive to improve our services solely on our own. Google Cloud helped us overcome these roadblocks with its advanced artificial intelligence (AI) and machine learning (ML) technologies,” he explains.

Today, Hi Translate deploys Translation API (/translate) as the backbone of its text translation tools. After moving to Google Cloud, the number of languages available on Hi Translate’s apps has increased from 66 to 108, including 80% of minor languages in India.

“Among all the products we evaluated, Cloud Translation API provides the best translation quality and supports the most languages across our target markets,” says Yang. “We are therefore able to greatly optimize our translation applications and reach wider users.”

With Google Cloud, Hi Translate also managed to broaden the features of its translation apps. Using Cloud Translation API together with Speech-to-Text (/speech-to-text), the team successfully realized instant voice translation. In addition, the optical character recognition feature of Cloud Vision (/vision) allows Hi Translate to provide image translation in multiple languages simultaneously, which means users of Hi Translate’s apps can get the translation

of a document in various foreign languages simply by taking a picture.



Improving the user experience with stable connection and shorter response time

Before the deployment on Google Cloud, most of the error reports that Hi Translate received were related to connection issues, especially from regions with less advanced internet infrastructures. At that time, Hi Translate used to encounter one to two service outages every year. After each incident, the amount of its daily active users slumped by 10% to 20%. Thanks to the highly connected global network supporting all the Google Cloud products, the stability of Hi Translate's apps has considerably improved, with no downtime recorded and a 55% decrease in the number of error reports since the move.

Another advancement brought by Google Cloud is the shortened translation processing time. "The high connectivity of the Google Cloud network has helped us reduce the time required for our service to

generate translations from one second to less than 600 milliseconds. This means our users are now able to work, study, or communicate 40% more efficiently,” says Yang.

The positive feedback that Hi Translate received from its users after the migration to Google Cloud also reflects the overall enhancement of its applications. Between June and November 2020, the number of downloads of its two apps increased by 300%, while their ratings on Google Play both climbed from 4.0 to 4.5.

Accelerating app development with intuitive cloud products

Since its founding, Hi Translate has kept a lean team of 10 employees, largely due to the intuitiveness of Google Cloud tools that boosts work efficiency. “In the past, we would require one week to deploy our two apps in a new environment. We were surprised that moving to Google Cloud only took us seven hours,” details Yang.

He explains that the documentation of Google Cloud products is very easy to understand and does not require much technical know-how to harness these tools, which has significantly increased the team’s productivity. Leveraging [Cloud SDK \(/sdk\)](#) for its app development, Hi Translate is almost ready to launch its third product, a voice translation app, after working on Google Cloud for only a couple of months.

Additionally, as Google Cloud provides comprehensive backend support, Hi Translate is able to free up DevOps workforce for product development by reducing its maintenance team from three to one employee. The team can therefore focus on improving its services, without worrying about the stability issues of the underlying technology.

“For us,
deploying
Google Cloud is
more than
acquiring tools.

As its
technologies
advance, it
strengthens our
development
capabilities to
realize the ideas
that seemed
impossible
before, such as
real-time
speech-to-

speech
translation.”

—Li Yang, CEO, Hi
Translate

Innovating translation tools to provide faster services

To reach its mission of eliminating language barriers, Hi Translate has to keep innovating its products to provide faster translation services applicable to broader settings. With the constantly improving Google Cloud technologies, Hi Translate has found more possibilities to upgrade its services. The team is currently using a Google Cloud beta product that combines the APIs of Speech-to-Text, Translation, and Text-to-Speech (/text-to-speech) to develop a real-time speech-to-speech translation tool, which will allow people to talk face-to-face with each other in different languages.

“Building a useful real-time speech translation feature was not easy, because the response time was unbearably long, as it required three APIs to process the translation,” explains Yang. “By merging the three APIs, Google Cloud has not only helped us remove this technical obstacle, but also simplified our development process.”

Hi Translate is also considering deploying Cloud TPU (/tpu) to train ML models that can support instant translation of daily chats in colloquial

language, as well as a chatbot to help Hi Dictionary users to practice speaking. The company is also planning to make use of the Google ecosystem to further integrate its apps with Android smartphones.

“For us, deploying Google Cloud is more than acquiring tools. As its technologies advance, it strengthens our development capabilities to realize the ideas that seemed impossible before, such as real-time speech-to-speech translation. We are now confident that we can further break down the world’s language barriers by providing translation services as fast and accurate as possible for all conversation settings,” Yang says.